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MEASUREMENT
SENSITIVE

MIL-PRF-83779D
13 January 2025

SUPERSEDING
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16 December 2013

PERFORMANCE SPECIFICATION
WATER INDICATING PASTE

Inactive for new design after 21 October 2005

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers water indicating paste used to determine the water level in tanks of aviation jet fuels and marine diesel.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB OH 45433 or e-mailed to AFPET.PTPS@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6850

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-DTL-5624	-	Turbine Fuel, Aviation, Grade JP-5 (NATO F-44)
MIL-DTL-16884	-	Fuel, Naval Distillate
MIL-DTL-83133	-	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34) and NATO F-35
MIL-DTL-85470	-	Inhibitor, Icing, Fuel System, High Flash NATO Code Number S-1745

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS

Title 29, Part 1910	-	Occupational Safety and Health Standards

(Copies of this document are available online at https://www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D1655	-	Standard Specification for Aviation Turbine Fuels

(Copies of this document are available from https://www.astm.org.)

UK MINISTRY OF DEFENCE

Defence Standard 91-091	-	Turbine Fuel, Kerosene Type, Jet A-1; NATO Code: F-35; Joint Service Designation: AVTUR

(Copies of this document are available from https://sts.defencegateway.mod.uk/.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection in accordance with 4.1 (see 6.2).

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Hazardous Materials. The water indicating paste shall not contain carcinogenic or potentially carcinogenic constituents as defined under the Hazard Communication Standard 29 CFR 1910.1200.

3.4 Water Indication. The water indicating paste shall clearly indicate contact with water by a definite color change under the following conditions (see 4.3):

a. Ambient temperature range, -23 °C to 43 °C.

b. In water levels, containing from 0 percent to 60 percent fuel system icing inhibitor.

c. In water levels containing up to 1,000 parts per million chloride, by weight (ppmw).

3.5 Time for color change. The maximum time required for the water indicating paste to change color in contact with the water level shall be 30 seconds (See 4.3.3).

3.6 Storage stability. The compound shall remain stable and shall be completely suitable for use for one year from date of manufacture (See 4.4).

3.7 Markings. Each container of water indicating paste shall be clearly and legibly marked with the manufacturer's instructions for use. Instructions shall include specific directions for the most effective use of the particular water indicating paste, such as, but not limited to, the recommended thickness of application and approximate drying time before immersion as applicable.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).

b. Quality conformance inspection (see 4.2.2).

4.2 First article and quality conformance inspections.

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4.2.1 First article inspection. Preproduction samples consisting of twelve (12) jars or tubes of product, as applicable, shall be forwarded by the contractor, for performance of all examination and tests specified herein, to the designated laboratory (see 6.2 and 6.3). The product shall meet the requirements in 3.4 and 3.5.

4.2.2 Quality conformance inspection. Quality conformance inspections shall be accomplished by subjecting the product to all of the examinations and tests specified herein. The sample size shall be twelve (12) units per lot. The product shall meet the requirements in 3.4 and 3.5.

4.2.2.1 Place of inspection. Quality conformance inspection shall be performed at the source.

4.3 Quality control tests.

4.3.1 Test solutions. These solutions simulate a JP-8, Jet A or Jet A-1 tank containing high flash fuel system icing inhibitor (FSII), a JP-5 fuel tank containing FSII, and a naval distillate fuel tank, all having water bottoms containing salt water. These solutions are used in evaluating the effectiveness of water indicating paste in determining the water cut in fuel tanks containing JP-8, Jet A or Jet A-1/FSII, JP-5/FSII, and naval distillate fuel with salt water bottoms. Should JP-8 not be available as test fuel, refer to 4.6.

4.3.1.1 Solution I, JP-8, Jet A or Jet A-1/FSII. This mixture is made up of JP-8, Jet A, or Jet A-1 with a synthetic water bottom containing 60 percent fuel system icing inhibitor and 1000 ppmw sodium chloride.

Prepare as follows:

a. Add 3,500 mL of JP-8 (conforming to MIL-DTL-83133, NATO F-34) or Jet A or Jet A-1 (conforming to ASTM D1655) to a suitable container.

b. In a separate 750 mL beaker, add 0.5069 grams (g) reagent grade sodium chloride, to 200 mL distilled water. Stir to dissolve and add 300 mL FSII (conforming to MIL-DTL-85470, NATO S-1745).

c. Add the FSII/sodium chloride solution to the container of fuel, allowing the beaker to drain thoroughly. Do not rinse the beaker into the fuel container.

4.3.1.2 Solution II, JP-5/FSII. This mixture consists of JP-5 with a synthetic water bottom containing 60 percent FSII and 1000 ppmw sodium chloride. Prepare as follows:

a. Add 3,500 mL of JP-5 (conforming to MIL-DTL-5624, NATO F-44) to a suitable container.

b. In a separate 750 mL beaker, add 0.5069 g reagent grade sodium chloride to 200 mL distilled water, stir to dissolve, and add 300 mL FSII (conforming to MIL-DTL-85470, NATO S-1745).

c. Add the FSII/sodium chloride solution to the container of JP-5, allowing the beaker to drain thoroughly. Do not rinse the beaker into the JP-5 container.

4.3.1.3 Solution III, marine distillate fuel/salt water (water bottom). This mixture consists of marine distillate fuel with a synthetic water bottom containing 1,000 ppmw sodium chloride. Prepare as follows:

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a. Add 3,500 mL naval distillate fuel (conforming to MIL-DTL-16884, NATO F-76) to a suitable container.

b. In a separate 750 mL beaker, add 0.5000 g reagent grade sodium chloride to 500 mL distilled water, and stir to dissolve.

c. Add the salt solution to the container of marine distillate fuel, allowing the beaker to drain thoroughly. Do not rinse the beaker into the container of marine distillate fuel.

4.3.2 Test procedures.

4.3.2.1 Preparation of water indicating paste. Prepare water indicating paste for test as follows:

a. Chill six (6) tubes or jars of water indicating paste (whichever the contract or order specified) to -23 °C and maintain at this temperature for at least two hours.

b. Heat six (6) tubes or jars of water indicating paste (whichever the contract or order specified) to 43 °C and maintain at this temperature for at least two hours.

4.3.2.2 Preparation of solutions.

a. Chill Solution I (JP-8, Jet A or Jet A-1/FSII solution) and Solution II (JP-5/FSII) until the fuel phase is -7 °C.

b. Chill Solution III (naval distillate fuel/salt solution) until the fuel phase is 2 °C.

4.3.3 Testing water indicating paste.

4.3.3.1 Chilled paste, chilled Solution I (JP-8, Jet A or Jet A-1/FSII).

a. Remove an unused jar or tube of paste that has been chilled to -23 °C from the cold box and immediately spread the paste on a typical dip stick or rod, according to the instructions on the tube or jar.

b. Lower the stick or rod into Solution I, which has been chilled to -7 °C, through the fuel and into the water phase containing the FSII. Allow the rod or stick to remain in the water bottom for 30 ± 5 seconds. Raise the rod or stick into the fuel level and let it remain there for 60 ± 5 seconds. Remove from the fuel and observe the color change at the water cut.

c. The water cut shall be clearly identifiable by a distinct, sharp color change (see 3.4 and 3.5).

4.3.3.2 Chilled paste, chilled Solution II (JP-5/FSII).

a. Remove an unused tube or jar of paste from the cold box, and follow the same procedure as in paragraph 4.3.3.1, using Solution II chilled to -7 °C.

b. The water cut shall be clearly identifiable by a distinct, sharp color change (see 3.4 and 3.5).

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4.3.3.3 Chilled paste, chilled Solution III (marine distillate fuel/salt solution).

a. Remove an unused tube or jar or paste from the cold box, and follow the same procedure as in paragraph 4.3.3.1, using Solution III chilled to 2 °C.

b. The water cut shall be clearly identifiable by a distinct, sharp color change (see 3.4 and 3.5).

4.3.3.4 Heated paste, chilled Solution I (JP-8, Jet A or Jet A-1/FSII).

a. Remove an unused jar or tube of paste that has been heated to 43 °C from the oven and immediately spread the paste on a dip stick or rod according to instructions on the tube or jar.

b. Lower the stick or rod into Solution I, which has been chilled to -7 °C, through the fuel and into the water phase containing the FSII. Allow the rod or stick to remain in the water bottom for 30 ± 5 seconds. Raise the rod or stick into the fuel level and let it remain there for 60 ± 5 seconds. Remove from the fuel and observe the color change at the water cut.

c. The water cut shall be clearly identifiable by a distinct, sharp color change (see 3.4 and 3.5).

4.3.3.5 Heated paste, chilled Solution II (JP-5/FSII).

a. Remove an unused tube or jar of paste from the oven and follow the same procedure as in 4.3.3.4, using Solution II chilled to -7 °C.

b. The water cut shall be clearly identifiable by a distinct, sharp color change (see 3.4 and 3.5).

4.3.3.6 Heated paste, chilled Solution III (marine distillate fuel/salt solution).

a. Remove an unused tube or jar of paste from the oven and follow the same procedure as in 4.3.3.4, using Solution III chilled to 2 °C.

b. The water cut shall be clearly identifiable by a distinct, sharp color change (see 3.4 and 3.5).

4.3.3.7 Chilled paste, ambient temperature Solution I, Solution II, and Solution III.

a. Allow the three Solutions prepared in paragraph 4.3.1 (Solution I - JP-8, Jet A or Jet A-1/FSII, Solution II - JP-5/FSII, and Solution III - marine distillate fuel/salt solution) to reach ambient temperature.

b. Remove an unused jar or tube that has been chilled to -23 °C from the cold box and immediately spread the paste on a dip stick or rod according to instructions on the tube or jar.

c. Lower the stick or rod into Solution I through the fuel and into the water phase containing the FSII. Allow the rod or stick to remain in the water bottom for 30 ± 5 seconds. Raise the rod or stick into the fuel level and let it remain there for 60 ± 5 seconds. Remove from the fuel and observe the color change at the water cut.

d. The water cut shall be clearly identifiable by a distinct, sharp color change (see 3.4

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and 3.5).

e. Repeat 4.3.3.7(b) through 4.3.3.7(d) for ambient Solutions II and III.

4.3.3.8 Heated paste, ambient temperature Solution I, Solution II, and Solution III.

a. Allow the three Solutions prepared in paragraph 4.3.1 (Solution I - JP-8, Jet A or Jet A-1/FSII, Solution II - JP-5/FSII, and Solution III - marine distillate fuel/salt solution) to reach ambient temperature.

b. Remove an unused jar or tube that has been heated to 43 °C from the oven and immediately spread the paste on a dip stick or rod according to instructions on the tube or jar.

c. Lower the stick or rod into Solution I through the fuel and into the water phase containing the FSII. Allow the rod or stick to remain in the water bottom for 30 ± 5 seconds. Raise the rod or stick into the fuel level and let it remain there for 60 ± 5 seconds. Remove from the fuel and observe the color change at the water cut.

d. The water cut shall be clearly identifiable by a distinct, sharp color change (see 3.4 and 3.5).

e. Repeat 4.3.3.8(b) through 4.3.3.8(d) for ambient temperature Solutions II and III.

4.4 Storage stability. Product shall pass all specification requirements when tested one year from date of manufacture.

4.5 Analytical procedures. Unless otherwise specified, all samples shall be analyzed according to the procedures described in this document (see 6.2).

4.5.1 Documentation of procedures. Written procedures that detail the sampling and testing program and the analytical and measuring equipment calibration program shall be maintained.

4.6 Availability of JP-8 as test fuel. Should JP-8 not be available, one of the following can be used as suitable replacements.

a. NATO F-24, which is ASTM D1655 Grade Jet A additized per the requirements for JP-8 in MIL-DTL-83133.

b. A fully additized Jet A-1 meeting either Defence Standard 91-091 or ASTM D1655 additized per the requirements for JP-8 in MIL-DTL-83133.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Water indicating paste covered by this specification is intended to be applied to dip sticks, rods, plumb bobs, etc., used to determine depth of water (or water bottoms) in military aviation or marine diesel fuel storage tanks.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article inspection will be performed as specified in 4.2.1.

c. The name and address of the laboratory to which the representative product specimen is to be submitted (see 4.2 and 6.3).

d. When variation in sampling method is granted (see 4.3.2).

e. When a variation in analytical procedures is granted (see 4.5).

f. Packaging requirements (see 5.1).

g. Shelf-life requirements (see 3.6, 4.4, and 6.4).

h. Method of shipment.

6.3 Government laboratory. First article inspection samples for testing are to be submitted to AFPET/ PTPLA, 2430 C Street, Bldg. 70, Area B, Wright-Patterson AFB OH 45433.

6.4 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings IAW MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, Volume I, Shelf-Life Management Program: Program Administration; and Volume 2, Shelf-Life Management Program: Materiel Quality Control Storage Standards. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.5 Lot definition. By definition, a packaged lot is an indefinite number of unit packages of identical size and type, offered for acceptance and filled with a homogeneous mixture of material from one isolated container; or filled with a homogeneous mixture of material manufactured by a single plant run through the same processing equipment with no change in ingredient material.

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6.6 Subject term (key word) listing.

Aerospace
F-24
FSII
Fuel
Jet A
Jet A-1
JP-5
JP-8
Marine Diesel
Petroleum
Tank

6.7 Scalability. The test solutions may be scaled to decrease total sample volume to meet laboratory equipment capabilities provided the fuel/aqueous ratio is equivalent to the volumes found in 4.3.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – EA
Navy – SH
Air Force – 68
DLA – GS

Review activities:
Army – AT
Navy – AS, MC, SA, YD

Preparing activity:
Air Force – 68
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